

AE646 – STAGE 2 INTERIM EVALUATION

Operator Learning for Parametric Darcy Flow: FNO vs MLP Baseline

Evaluating spectral inductive biases against fully-connected architectures for continuous PDE mappings

TEAM PINNACLES

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COURSE & DATE

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2D Parametric Darcy Flow Governing Equation

Governing PDE System

$$-\nabla \cdot (\kappa(x, y) \nabla u(x, y)) = f(x, y)$$

Domain: $(x, y) \in [0, 1]^2$ with homogeneous Dirichlet BC $u = 0$.

Permeability: Piecewise-constant field $\kappa \in \{0.1, 1.0\}$.

Forcing: Constant unit forcing $f = 1.0$.



Computational Bottleneck

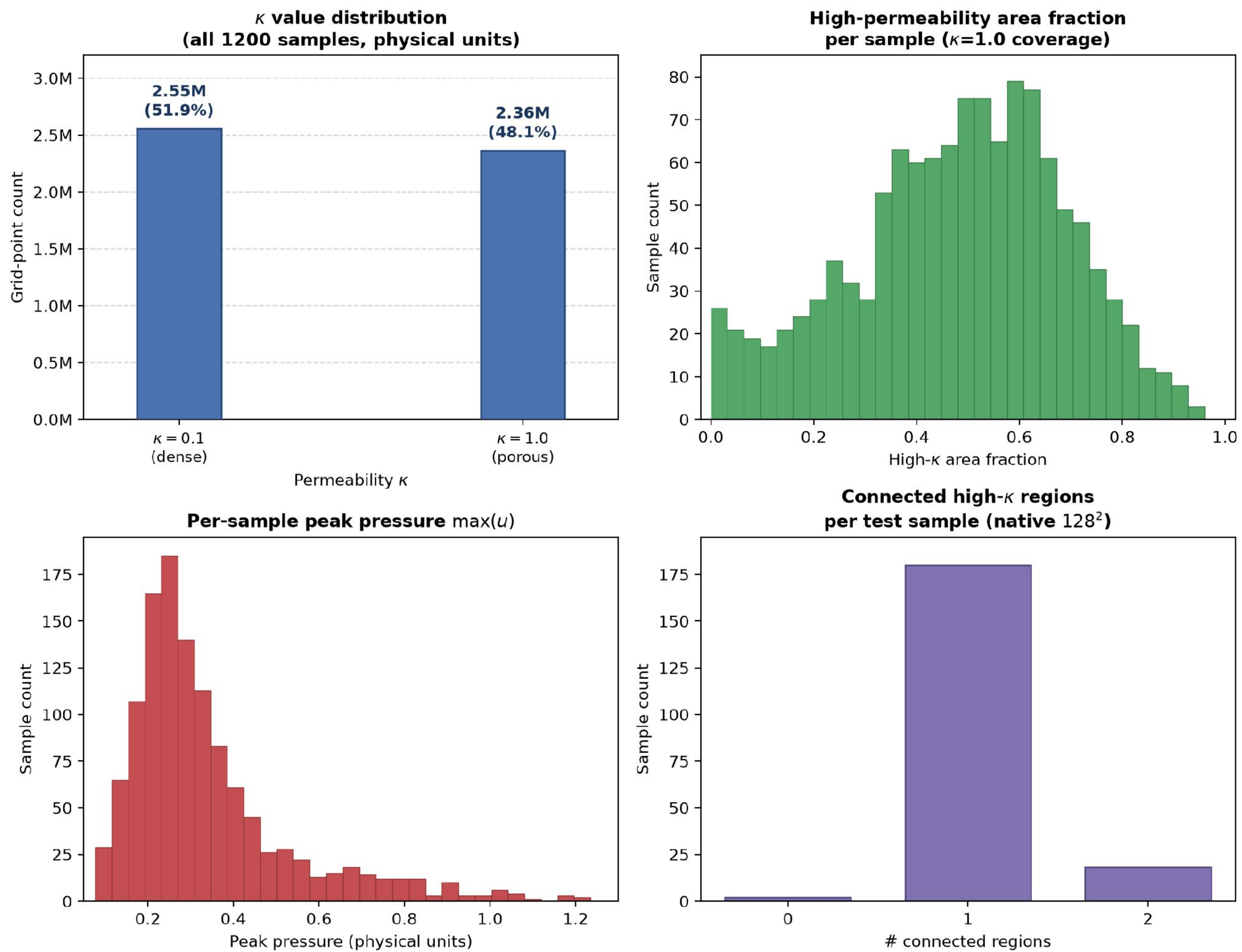
FDM / FEM Solvers: Re-solving full PDE for every new coefficient field is prohibitively slow for UQ, control, and inverse design.



Operator Learning Goal

Neural Operator Mapping: Learn continuous mapping $G : \kappa \mapsto u$ to amortize solve cost into rapid single-pass neural evaluation.

PDEBench Dataset & Statistical Characterisation



Data Source & Splits

10,000 samples at 128 × 128 downsampled to 64 × 64. 900 train / 100 val / 200 held-out test split.

Bimodal Permeability

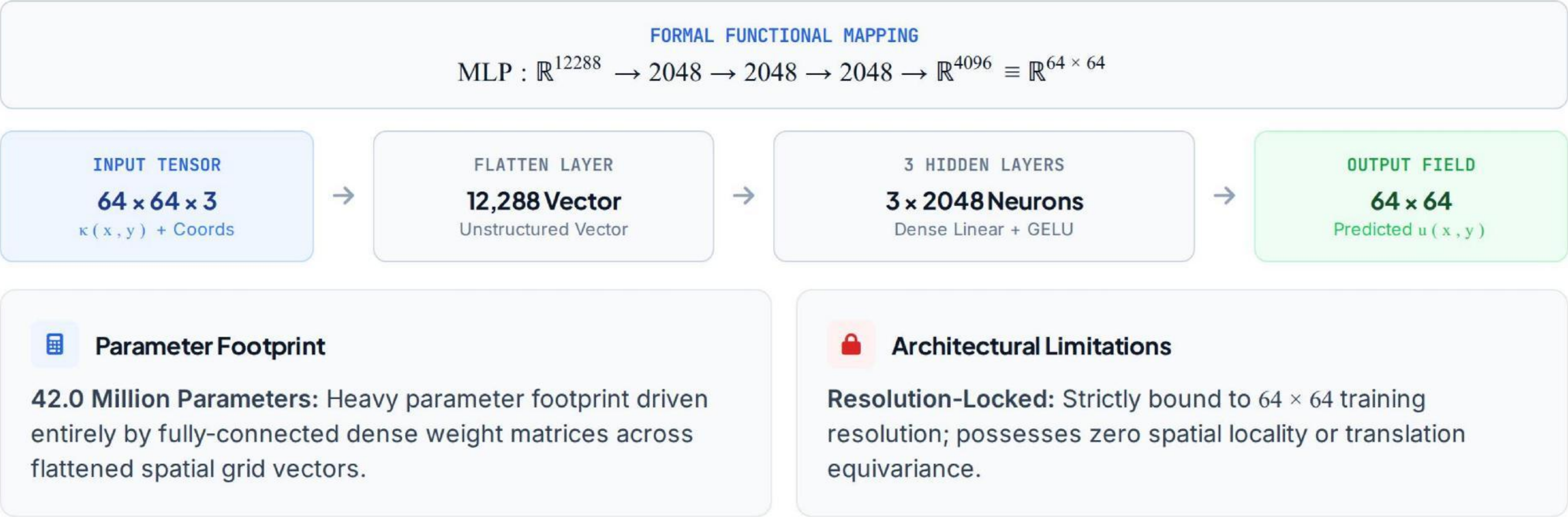
Strictly discrete $\kappa \in \{0.1, 1.0\}$. Peak pressure ranges 0.1 to >1.2 physical units.

Geometric Character

180/200 single-blob fields, 18 multi-region fields, 2 degenerate uniform ($\kappa \equiv 0.1$) fields.

Data Pipeline Status: Preprocessing, normalisation, and non-overlapping splits verified. 200 held-out test fields kept at native 128 × 128 resolution for zero-shot testing.

MLP Baseline Architecture & Mathematical Mapping



Preprocessing Ablation Study (MLP Baseline)

Evaluated on 200 test samples across 3 independent seeds per variant to confirm transformation choices.

Preprocessing Variant	Inputs	Targets	Test Mean Rel. L ₂ Error
Reference Pipeline	Standardised + Coords (64 × 64 × 3)	Standardised Pressure	0.0842 ± 0.0002
No Coordinate Channels	Standardised κ Only (64 × 64 × 1)	Standardised Pressure	0.0843 ± 0.0011
No Input Normalisation	Raw κ + Coords	Standardised Pressure	0.0864 ± 0.0018
No Target Normalisation	Standardised + Coords	Raw Pressure (Physical)	0.0917 ± 0.0012

Target Normalisation Essential: Omitting pressure standardization causes a **+9% relative error spike**, well outside seed variance.

Coordinate Channels Preserved: Flattened MLPs fix pixel positions implicitly, but coordinate channels maintain identical input structure across FNO comparisons.

MLP Baseline Ablation: Depth & Optimizer Impact

Evaluated on 200 held-out test samples to determine optimal depth and optimizer configurations.

Configuration	Parameters	Optimizer	Mean Rel. L ₂	Median Rel. L ₂
1 Hidden Layer (2048)	33.6 M	AdamW	0.1008	0.0892
2 Hidden Layers (2048 × 2)	37.8 M	AdamW	0.0796	0.0649
3 Hidden Layers (2048 × 3) (Final Baseline)	42.0 M	AdamW	0.0857	0.0700
3 Hidden Layers (2048 × 3)	42.0 M	SGD + Momentum	0.1209	0.1091

Key Ablation Takeaways: SGD optimizer severely underperforms AdamW (0.1209 vs 0.0857). Depth beyond 2 layers yields no significant gain (gap within ~0.004 run noise floor); 3-layer baseline retained as a fair benchmark.

Fourier Neural Operator (FNO2D) Architecture

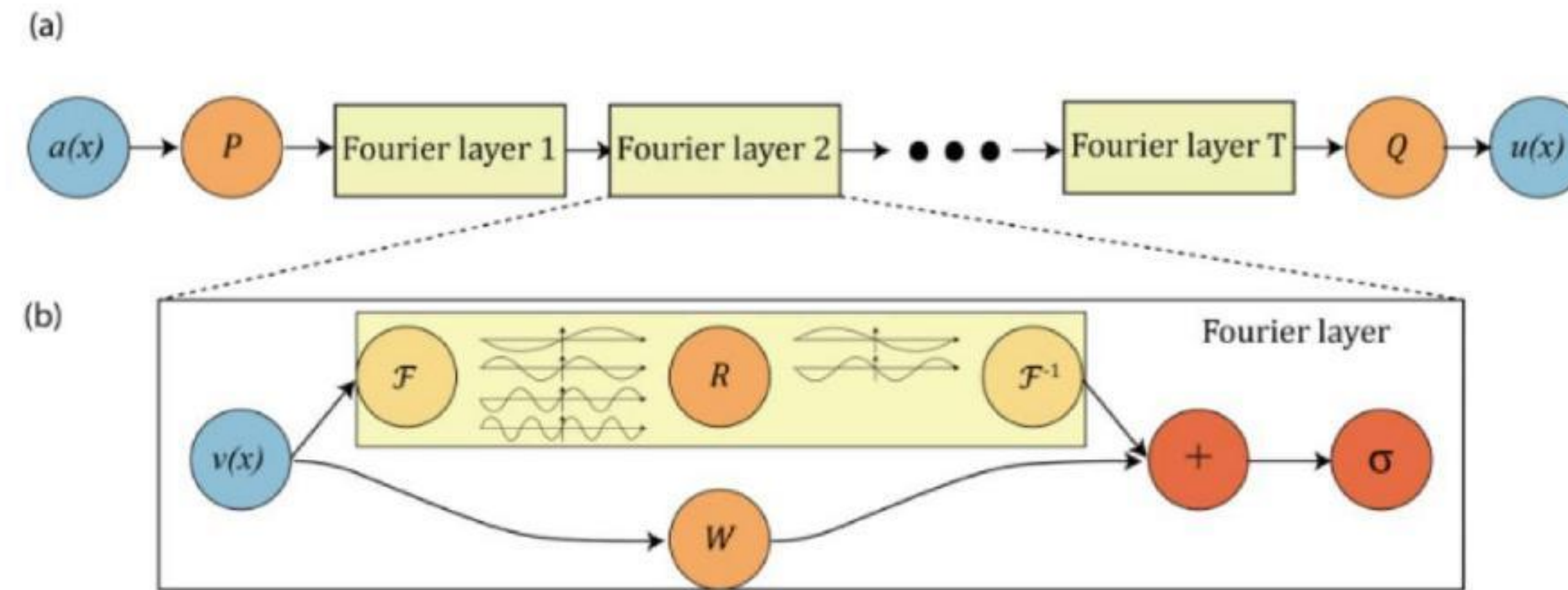


Figure 3: Architecture of the Fourier Neural Operator used in this work.

⚡ 9× Parameter Reduction

4.7 Million Parameters: Trainable weights parameterized in low-frequency Fourier domain rather than on high-dimensional grid matrices.

🔲 Resolution Invariance

Discretisation-Invariant: Continuous operator learning enables direct zero-shot evaluation on arbitrary grid resolutions.

Shared Training Protocol & Optimization Strategy

Loss Function & Metric

Loss: MSE on standardised pressure fields.

Reported Metric: Relative L_2 error $\| \hat{u} - u \| / \| u \|$ evaluated strictly in physical units.

Optimizer & Decay

AdamW Optimizer: Decoupled weight decay of 10^{-4} and peak learning rate of 10^{-3} across all model variants.

Step Decay & Warmup Schedule

Adaptive Decay Protocol: 5-epoch linear warm-up followed by StepLR reduction ($\gamma = 0.5$ every 25 epochs) for smooth optimization convergence.

Budget & Checkpointing

Full Epoch Budget: Batch size 16 for 100 full epochs. Retained best-validation checkpoint (Epoch 92 for FNO, Epoch 99 for MLP).

Quantitative Benchmark: FNO vs MLP Baseline

ERROR REDUCTION

36%

Lower Mean Relative L₂ Error

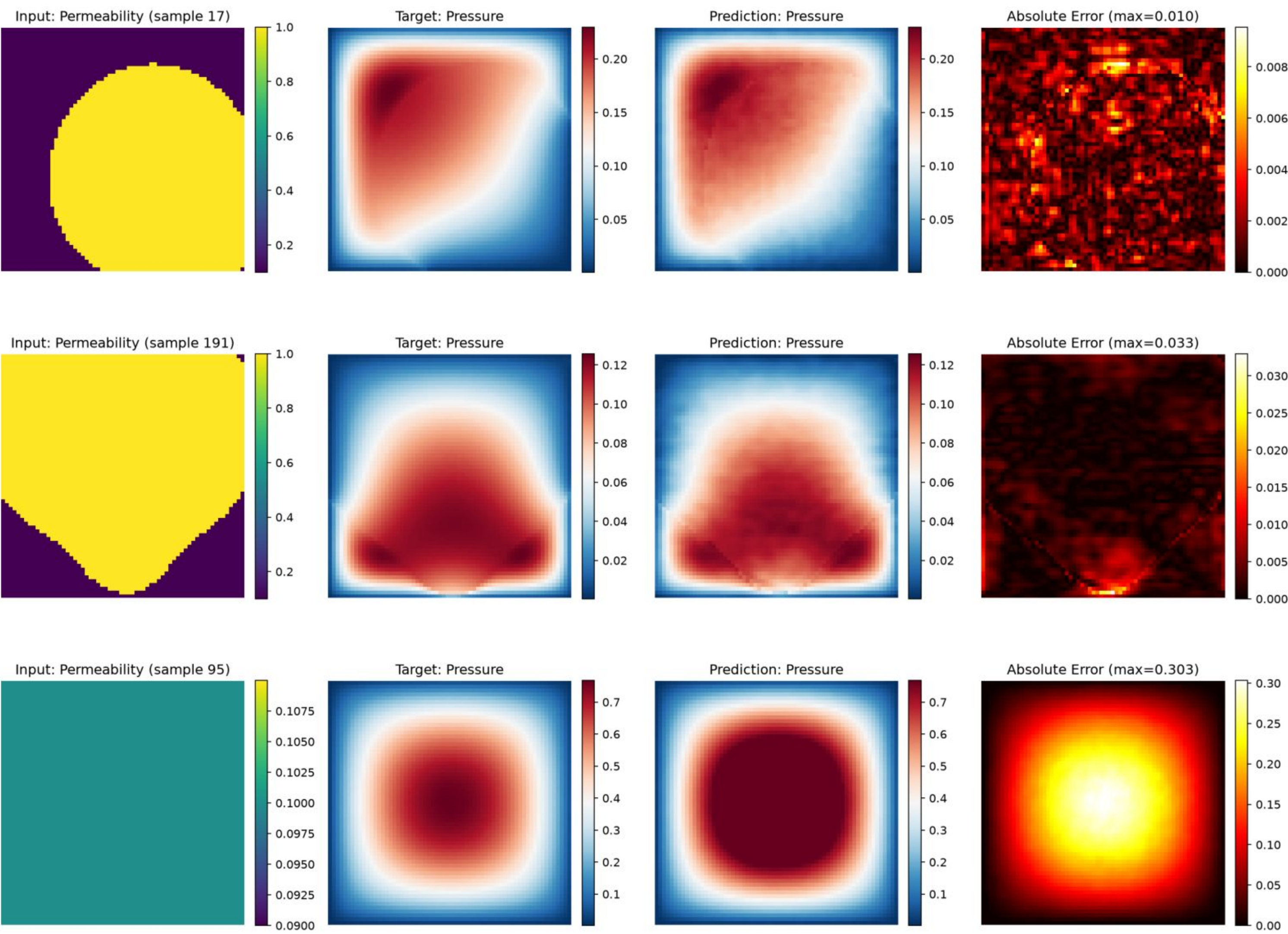
PARAMETER EFFICIENCY

9x

Fewer Parameters (4.7M vs 42.0M)

Model Architecture	Params	Mean Rel. L ₂	Median Rel. L ₂	Std Dev	Min Error	Max Error
MLP Baseline	42.0 M	0.0820	0.0695	0.0456	0.0325	0.3367
Fourier Neural Operator (FNO)	4.7 M	0.0521	0.0398	0.0451	0.0155	0.3467

Sample Predictions & Edge Case Analysis



BEST CASE (SAMPLE 17)
Rel. L_2 Error: 0.016
High accuracy across smooth interior pressure profiles and high-contrast boundary edges.

MEDIAN CASE (SAMPLE 191)
Rel. L_2 Error: 0.040
Accurate physical recovery of pressure gradient orientation for standard geometries.

DEGENERATE CASE (SAMPLE 95)
Rel. L_2 Error: 0.347
Uniform $\kappa \equiv 0.1$ field causes FNO to over-predict pressure peak amplitude.

Failure Mode Insight: FNO max error is driven by rare degenerate uniform inputs (2/200 test cases) rather than complex geometric fields. Reported openly for complete transparency.

Stage 3 Planned Evaluation & Research Objectives

1. FNO SCALING

Width & Modes

Map accuracy vs parameter trade-offs across larger modes and hidden channels.

2. ZERO-SHOT

Resolution Transfer

Evaluate 64×64 -trained FNO zero-shot on native 128×128 grid test fields.

3. LATENCY

Cost Benchmarking

Benchmark inference speedup against classical reference FDM numerical solvers.

4. GEOMETRY

Physical Analysis

Correlate per-sample loss with permeability interface length and region geometry.

SUMMARY & OUTLOOK

Operator Learning for Parametric Darcy Flow

Data Pipeline

PDEBench 10k dataset pipeline complete with verified preprocessing ablations.

Key Finding

FNO achieves 36% error reduction with 9× fewer parameters than tuned MLP.

Thank You — Questions & Discussion Welcome

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